

Notes: Davis and Haltiwanger (QJE, 1992)
Gross Job Creation, Gross Job Destruction, and Employment Reallocation

Contents

1	Big picture	2
2	Research question	2
3	Incremental contribution of the paper	2
3.1	First contribution: direct establishment-level measurement	2
3.2	Second contribution: gross job flows rather than net flows	3
3.3	Third contribution: link between job reallocation and worker reallocation	3
3.4	Fourth contribution: decomposition of sources of heterogeneity	3
3.5	Fifth contribution: cyclical behavior of gross job flows	3
4	Data and measurement	3
4.1	Data source: the Longitudinal Research Datafile	3
4.2	Growth-rate measure	4
4.3	Gross job creation	4
4.4	Gross job destruction	4
4.5	Net employment growth and job reallocation	5
4.6	Lower bound on worker reallocation required by job reallocation	5
5	Main empirical facts	5
5.1	Fact 1: gross flows are large	5
5.2	Fact 2: simultaneous creation and destruction occurs every year	5
5.3	Fact 3: job reallocation is pervasive across industries	5
5.4	Fact 4: job reallocation explains a large share of worker reallocation	6
5.5	Fact 5: job creation and destruction are persistent	6
5.6	Fact 6: job destruction is concentrated	6
6	Sources of job reallocation	6
6.1	Passive learning about initial conditions	6
6.2	Between-sector shifts	7
6.3	Interpretation	7

7	Cyclical behavior of job reallocation	7
7.1	Countercyclicality	7
7.2	Aggregate, sectoral, and idiosyncratic components	7
7.3	Key result on time-series decomposition	8
7.4	Heterogeneity by plant type	8
8	Identification and empirical design	8
9	Relationship to theory	8
9.1	Passive learning and selection	9
9.2	Vintage and technological replacement	9
9.3	Idiosyncratic plant-level shocks	9
9.4	Macroeconomic models	9
10	Extracted figures and tables from the paper	10
10.1	Figures Ia and Ib: Unweighted and Size-Weighted Growth Rate Distributions	10
10.2	Tables I and II: Net and Gross Rates by Year and Industry	11
10.3	Figure II and Table III: Growth-Rate Partition and Persistence	12
10.4	Tables IV and V: Plant Characteristics and Passive Learning	13
10.5	Tables VI and VII: Between-Group Shifts and Time-Series Variance Decomposition .	14
10.6	Tables VIII and IX: Cyclical Behavior and Regression Evidence	15
11	Key formulas and intuition	15
11.1	Why use the symmetric growth rate?	15
11.2	Why does <i>SUM</i> matter?	16
11.3	Why is excess reallocation important?	16
12	Main conclusions	16
12.1	Core conclusion	16
12.2	Economic intuition	16
12.3	Incremental contribution revisited	16
12.4	Limitations	16
12.5	Takeaway	17
13	One-sentence summary	17

1 Big picture

This note summarizes Davis and Haltiwanger (1992), a foundational paper that constructs establishment-level measures of gross job creation, gross job destruction, and job reallocation in U.S. manufacturing. The paper shifts attention from net employment changes to the much larger gross flows underneath those net changes. Its core insight is that even when total manufacturing employment changes only modestly, many plants are expanding while many others are contracting. Therefore, labor-market dynamics are not well described by a representative firm or even by homogeneous firms within broad sectors.

The paper uses the Longitudinal Research Datafile (LRD), which contains approximately 860,000 annual observations on roughly 160,000 U.S. manufacturing establishments over 1972-1986. This plant-level panel allows the authors to measure the reallocation of jobs across plants, not just the reallocation of workers across labor-market states.

The main empirical finding is simple but powerful: annual gross job creation averages about 9.2 percent of manufacturing employment, while annual gross job destruction averages about 11.3 percent. Their sum, the job reallocation rate, averages about 20.5 percent. In other words, each year roughly one fifth of manufacturing jobs are either newly created at expanding or entering plants, or destroyed at contracting or exiting plants.

2 Research question

The paper asks four closely related questions.

1. How large are gross job creation and gross job destruction in U.S. manufacturing?
2. How much worker reallocation can be explained by the reallocation of employment opportunities across plants?
3. What sources of plant-level heterogeneity generate simultaneous job creation and destruction within narrowly defined sectors?
4. Why does job reallocation fluctuate countercyclically over the business cycle?

The key conceptual move is to separate *net employment growth* from *gross job flows*. A sector may lose 2 percent of jobs on net while simultaneously creating 9 percent and destroying 11 percent. The net number hides the gross reallocation process.

3 Incremental contribution of the paper

The incremental contribution has five parts.

3.1 First contribution: direct establishment-level measurement

Before this paper, many studies documented large worker flows using household data, such as flows among employment, unemployment, and nonparticipation. However, household data cannot

reveal whether workers move because jobs themselves disappear and appear elsewhere, or because workers sort and re-sort across a relatively fixed set of jobs. Davis and Haltiwanger use longitudinal establishment data to measure job flows directly.

3.2 Second contribution: gross job flows rather than net flows

The paper shows that net employment changes are a small residual of two large gross flows. The main equation is:

$$NET_t = POS_t - NEG_t,$$

where POS_t is gross job creation and NEG_t is gross job destruction. The magnitude of $POS_t + NEG_t$ is far larger than $|NET_t|$.

3.3 Third contribution: link between job reallocation and worker reallocation

The authors calculate bounds on how much worker mobility is required to accommodate job reallocation. They estimate that job reallocation accounts for 35–56 percent of total worker reallocation between employers or between employment and joblessness. This makes job reallocation a central driver of labor-market mobility.

3.4 Fourth contribution: decomposition of sources of heterogeneity

The paper evaluates two broad explanations for simultaneous job creation and destruction: passive learning about plant-level efficiency, and sectoral shifts across industries, regions, size groups, age groups, and ownership types. Passive learning explains only 11–13 percent of total job reallocation, while between-sector shifts explain little of excess reallocation. The paper therefore argues that plant-level heterogeneity within narrowly defined sectors is fundamental.

3.5 Fifth contribution: cyclical behavior of gross job flows

The paper documents that job reallocation is countercyclical. Job destruction is more volatile than job creation, and recessions are associated with intensified reallocation. Importantly, the countercyclicality is driven mainly by older, larger, and multi-unit plants, not by young, small, or single-unit plants.

4 Data and measurement

4.1 Data source: the Longitudinal Research Datafile

The empirical analysis uses the Longitudinal Research Datafile, a U.S. Census Bureau establishment-level panel for manufacturing. The LRD covers all U.S. manufacturing establishments with at least five employees in its sampling frame, and these establishments account for about 99 percent of manufacturing employment. Annual data are available from 1972 to 1986.

The data have several advantages:

- Establishments can be followed longitudinally.
- The sampling frame is broad and probability based.
- Plant births and deaths are incorporated.
- Establishments are distinguished from firms.
- Ownership transfers are distinguished from true births and deaths.

The paper generally excludes the first year of each Annual Survey of Manufactures panel, because linking observations across panels is less reliable in those years. This is why many tables omit 1974, 1979, and 1984.

4.2 Growth-rate measure

For establishment e at time t , size is defined as average employment across two consecutive years:

$$x_{et} = \frac{EMP_{et} + EMP_{e,t-1}}{2}.$$

The establishment growth rate is:

$$g_{et} = \frac{EMP_{et} - EMP_{e,t-1}}{x_{et}}.$$

This growth rate lies in $[-2, 2]$. A death has $g = -2$, and a birth has $g = 2$. This symmetric growth-rate measure treats births, deaths, expansions, and contractions in one common framework.

4.3 Gross job creation

Gross job creation in sector s at time t is the employment-weighted sum of positive establishment growth rates:

$$POS_{st} = \sum_{e \in E_{st}, g_{et} > 0} \frac{x_{et}}{X_{st}} g_{et}.$$

It measures the fraction of sector employment created at expanding and entering establishments.

4.4 Gross job destruction

Gross job destruction is the employment-weighted sum of absolute negative growth rates:

$$NEG_{st} = \sum_{e \in E_{st}, g_{et} < 0} \frac{x_{et}}{X_{st}} |g_{et}|.$$

It measures the fraction of sector employment destroyed at shrinking and exiting establishments.

4.5 Net employment growth and job reallocation

Net employment growth is:

$$NET_{st} = POS_{st} - NEG_{st}.$$

Gross job reallocation is:

$$SUM_{st} = POS_{st} + NEG_{st}.$$

The central empirical object is SUM_{st} , because it measures how rapidly employment positions are reshuffled across establishments.

4.6 Lower bound on worker reallocation required by job reallocation

The authors define:

$$MAX_{st} = \max\{POS_{st}, NEG_{st}\}.$$

This is a lower bound on the worker movement required to accommodate job reallocation, because at least the larger of newly created jobs and destroyed jobs must be matched by workers moving across jobs or employment statuses.

5 Main empirical facts

5.1 Fact 1: gross flows are large

Across U.S. manufacturing, annual gross job creation averages 9.2 percent and gross job destruction averages 11.3 percent. Therefore, job reallocation averages 20.5 percent per year. These magnitudes imply that the labor market is characterized by substantial ongoing reshuffling, even within a single broad sector.

5.2 Fact 2: simultaneous creation and destruction occurs every year

Even in expansion years, job destruction remains large. For example, in 1973 manufacturing employment grew by 7.1 percent net, but job destruction was still 6.1 percent. Similarly, in recessionary 1975, employment fell by 10 percent net, but job creation was still 6.7 percent. This means expansions do not simply create jobs everywhere, and recessions do not simply destroy jobs everywhere.

5.3 Fact 3: job reallocation is pervasive across industries

Every two-digit manufacturing industry exhibits significant gross job creation and destruction. The job reallocation rate ranges from about 14 percent in Tobacco to nearly 29 percent in Lumber and Wood Products. Therefore, job reallocation is not a narrow phenomenon concentrated in a few volatile industries.

5.4 Fact 4: job reallocation explains a large share of worker reallocation

Using CPS worker-flow data and LRD job-flow data, the authors calculate that 35–56 percent of total worker reallocation is required to accommodate shifts in employment opportunities across work sites. Job reallocation is therefore not merely a plant-level accounting concept; it is an important driver of individual worker mobility.

5.5 Fact 5: job creation and destruction are persistent

The average one-year persistence rate is 68 percent for newly created jobs and 81 percent for newly destroyed jobs. The persistence of job destruction is especially high. This rules out the interpretation that most annual job destruction reflects temporary layoffs followed by recall.

5.6 Fact 6: job destruction is concentrated

Only 23 percent of job destruction is accounted for by establishments that shrink by less than 20 percent in a year. A large fraction of destroyed jobs comes from establishments with sharp contractions or deaths. Therefore, the bulk of job destruction cannot be accommodated simply by normal worker attrition.

6 Sources of job reallocation

6.1 Passive learning about initial conditions

The paper evaluates models in which young plants learn about their own productivity or cost parameters over time. Inefficient plants shrink or exit, while efficient plants expand. Such models naturally predict high job reallocation among young plants and a decline in reallocation with plant age.

The authors estimate how much total job reallocation can be explained by this mechanism. They assume that by age n plants have learned their underlying efficiency, so job reallocation among mature plants represents a base rate unrelated to passive learning. The estimated contribution of learning is:

$$P = \left[\sum_{a < n} \frac{x(a)}{X} (r(a) - r(\text{age} \geq n)) - \max\{g, 0\} \right] / r,$$

where $r(a)$ is the job reallocation rate for age group a , $r(\text{age} \geq n)$ is the mature-plant base rate, g is long-run net growth, and r is overall reallocation.

The result is that passive learning explains only 11–13 percent of total job reallocation. This is because most manufacturing employment is located at plants older than six years, and these mature plants still exhibit high reallocation rates.

6.2 Between-sector shifts

The paper also asks whether excess job reallocation can be explained by employment shifts across sectors. Excess job reallocation is:

$$SUM - |NET|.$$

If sectoral shocks were the main source of reallocation, then employment shifts among industries, regions, size groups, or age groups should explain a large share of excess reallocation.

The authors decompose excess reallocation into between-sector and within-sector components. The between-sector component is:

$$\sum_s |\Delta E_s| - |\Delta E|.$$

The within-sector component is:

$$\sum_s (SUM_s - |NET_s|).$$

The striking result is that between-sector shifts explain very little. Employment shifts among two-digit industries, regions, age groups, size groups, and ownership types account for almost none of excess job reallocation. Even 450 four-digit industries account for only 12 percent of excess job reallocation.

6.3 Interpretation

The failure of passive learning and between-sector shifts to explain the overall magnitude of reallocation implies that job reallocation is primarily a within-sector, plant-level heterogeneity phenomenon. Any successful theory must explain why mature plants in narrowly defined industries simultaneously expand and contract at high rates.

7 Cyclical behavior of job reallocation

7.1 Countercyclicality

Job reallocation is countercyclical. The correlation between net employment growth and job reallocation is about -0.57 . During downturns, destruction rises more than creation falls, causing total reallocation to increase.

7.2 Aggregate, sectoral, and idiosyncratic components

The paper uses the decomposition:

$$g_{et} = g_{et}^{ST} + g_{st} + g_t,$$

where g_t is the aggregate manufacturing growth rate, g_{st} is the sector-time effect, and g_{et}^{ST} is the residual idiosyncratic plant-level component.

By removing aggregate and sectoral mean effects, the authors construct adjusted job-flow measures. If time variation in reallocation were caused by aggregate shifts or sectoral mean responses, adjusted reallocation should have little variation. Instead, the paper finds that idiosyncratic effects account for most time variation in job reallocation.

7.3 Key result on time-series decomposition

Aggregate and sectoral mean effects explain only 4–20 percent of the time variation in job reallocation. Idiosyncratic plant-level effects account for 80 percent or more. This is a central result: recessions are not merely periods in which all plants shift downward together; they are periods when plant-level employment changes become more dispersed.

7.4 Heterogeneity by plant type

The countercyclical behavior of reallocation is much stronger among:

- older plants,
- larger plants,
- multi-unit plants,
- durable-goods industries.

By contrast, job reallocation among young, small, and single-unit plants shows little systematic relationship with the cycle.

8 Identification and empirical design

This is primarily a measurement and decomposition paper rather than a causal identification paper. The empirical design relies on high-quality longitudinal establishment data and accounting identities that connect plant-level employment changes to aggregate job-flow measures.

The credibility of the paper comes from:

- comprehensive coverage of U.S. manufacturing establishments;
- annual longitudinal tracking of plants;
- separate treatment of births, deaths, expansions, and contractions;
- transparent definitions of *POS*, *NEG*, *NET*, *SUM*, and *MAX*;
- decomposition methods that require few behavioral assumptions.

The passive-learning calculation does require assumptions: learning is completed by age four or six, and non-learning sources of reallocation are age neutral after correcting for net growth. The sectoral-shift decomposition is closer to an accounting identity.

9 Relationship to theory

The paper evaluates several classes of theories.

9.1 Passive learning and selection

Jovanovic-style models predict high job reallocation among young plants as firms learn about their efficiency. The paper finds strong support for this mechanism in explaining the age gradient and cross-sectoral variation, but not the overall level of reallocation.

9.2 Vintage and technological replacement

Models of technological replacement predict simultaneous destruction at old plants and creation at new plants. These models are consistent with some facts but must also explain why mature plants continue to experience high reallocation.

9.3 Idiosyncratic plant-level shocks

The findings strongly support models with large idiosyncratic plant-level disturbances. These shocks may reflect plant-specific costs, demand shifts, technology changes, management differences, or local conditions.

9.4 Macroeconomic models

The paper rules out several common macro frameworks:

- representative-firm models;
- models with homogeneous firms within sectors;
- sectoral-shift models that explain cycles only through mean sectoral responses;
- models where idiosyncratic shocks are orthogonal to the business cycle.

A successful macro model must generate both simultaneous job creation and destruction within narrow sectors and countercyclical reallocation intensity.

10 Extracted figures and tables from the paper

10.1 Figures Ia and Ib: Unweighted and Size-Weighted Growth Rate Distributions

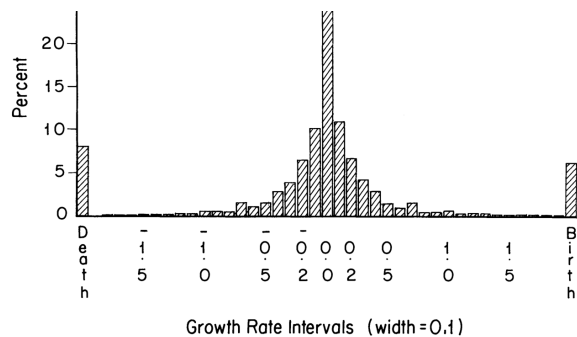


FIGURE Ia
Unweighted Growth Rate Distribution

ments in the empirical analysis. The g measure is monotonically related to the conventional growth rate measure, and the two measures are approximately equal for small growth rates.⁵

Figures Ia and Ib plot frequency distributions for the establishment growth rate observations in our eleven-year sample. Figure Ia depicts the shape of the empirical density over the 677,000

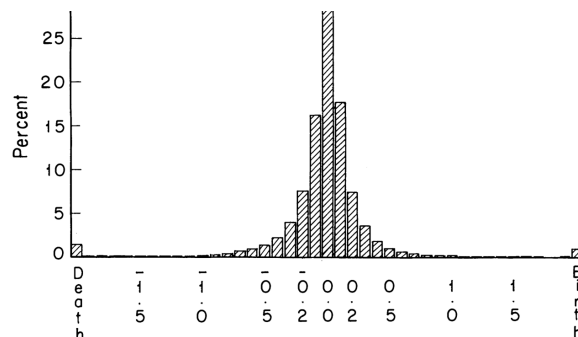


FIGURE Ib
Size-Weighted Growth Rate Distribution

tails. On a size-weighted basis, 29 percent of the annual growth rate observations fall in the interval $(-0.05, 0.05)$, and 63 percent fall in the interval $(-0.15, 0.15)$. Births and deaths account for only 0.4 percent of all size-weighted growth rate observations. But

(a) Figure Ia: Unweighted Growth Rate Distribution

(b) Figure Ib: Size-Weighted Growth Rate Distribution

Read: Figure Ia shows the unweighted distribution of establishment growth rates, while Figure Ib weights establishments by size. The unweighted distribution has large mass at births and deaths, because many plants enter or exit. The size-weighted distribution is more concentrated near zero, because small establishments account for a large share of entry, exit, and extreme growth observations.

Economic message: The figures show that plant-level employment growth is highly heterogeneous. Many establishments have small employment changes, but there is also meaningful mass in large expansions, large contractions, births, and deaths. Size weighting matters because extreme growth rates are more common among small plants.

10.2 Tables I and II: Net and Gross Rates by Year and Industry

QUARTERLY JOURNAL OF ECONOMICS

TABLE I
NET AND GROSS RATES BY YEAR, MANUFACTURING SECTOR

Year	POS_t	NEG_t	NET_t^a	SUM_t	MAX_t
1973	0.132	0.061	0.071	0.194	0.133
1975	0.067	0.166	-0.100	0.233	0.166
1976	0.113	0.096	0.017	0.209	0.122
1977	0.112	0.096	0.018	0.206	0.117
1978	0.116	0.075	0.041	0.191	0.117
1980	0.080	0.093	-0.012	0.173	0.102
1981	0.070	0.118	-0.049	0.188	0.119
1982	0.064	0.152	-0.087	0.216	0.152
1983	0.086	0.142	-0.056	0.227	0.143
1985	0.084	0.117	-0.033	0.201	0.121
1986	0.088	0.132	-0.044	0.220	0.133
Pearson correlations: ^b					
$\rho(POS_t, NEG_t) = -0.864 (0.001)$			$\rho(NET_t, SUM_t) = -0.565 (0.07)$		

(a) Table I: Net and Gross Rates by Year, Manufacturing Sector

Table II presents average annual net and gross job flow measures for the manufacturing sector and each two-digit industry. The industry figures are X_t -weighted averages of the eleven annual industry observations, and the figures for the manufacturing sector are X_t -weighted averages of the industry figures.

Employment contracted in every two-digit manufacturing industry over the sample period (1972–1986, exclusive of 1974, 1979, and 1984). Annual net contraction rates range from 0.2

TABLE II
NET AND GROSS RATES BY INDUSTRY, SIZE-WEIGHTED AVERAGES^a

Industry:	POS	NEG	NET	SUM	MAX	$SUM - NET $
Food (20)	0.089	0.104	-0.015	0.193	0.108	0.169
Tobacco (21)	0.058	0.082	-0.024	0.140	0.090	0.098
Textile (22)	0.074	0.110	-0.036	0.185	0.124	0.121
Apparel (23)	0.116	0.156	-0.040	0.272	0.168	0.207
Lumber (24)	0.129	0.160	-0.031	0.288	0.188	0.202
Furniture (25)	0.101	0.121	-0.019	0.222	0.143	0.157
Paper (26)	0.063	0.078	-0.015	0.141	0.089	0.105
Printing (27)	0.091	0.087	-0.004	0.178	0.099	0.158
Chemicals (28)	0.068	0.080	-0.013	0.148	0.089	0.118
Petroleum (29)	0.066	0.091	-0.025	0.157	0.100	0.114
Rubber (30)	0.107	0.118	-0.011	0.225	0.143	0.163
Leather (31)	0.091	0.144	-0.053	0.235	0.152	0.166
Stone, clay, and glass (32)	0.093	0.123	-0.031	0.216	0.136	0.160
Primary metals (33)	0.059	0.114	-0.054	0.173	0.126	0.094
Fabricated metals (34)	0.095	0.120	-0.025	0.215	0.137	0.156
Nonelectric machinery (35)	0.096	0.121	-0.025	0.217	0.141	0.152
Electric machinery (36)	0.097	0.109	-0.011	0.206	0.130	0.152
Transportation (37)	0.094	0.099	-0.006	0.193	0.123	0.140
Instruments (38)	0.093	0.093	-0.002	0.186	0.112	0.149
Miscellaneous (39)	0.108	0.145	-0.037	0.253	0.156	0.193
Total manufacturing	0.092	0.113	-0.021	0.205	0.129	0.152

(b) Table II: Net and Gross Rates by Industry, Size-Weighted Averages

Read: Table I reports annual job creation, destruction, net growth, total reallocation, and the lower bound on worker reallocation. Table II reports the same measures by two-digit industry. The main recurring pattern is that both POS and NEG are large in every year and industry.

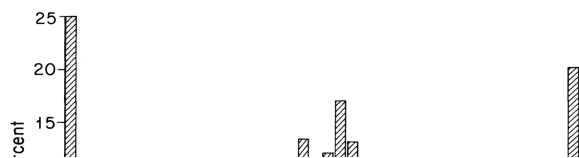
Economic message: Net employment growth is a residual. Large creation and destruction happen simultaneously, even during expansions and contractions. This fact motivates the whole job-flow literature.

10.3 Figure II and Table III: Growth-Rate Partition and Persistence

D. Concentration and Persistence

The high rates of job reallocation reported in Tables I and II prompt two further factual questions. First, what role do plant births and deaths play in the creation and destruction of jobs? Or, to restate the question in a more general way, how are job creation and destruction distributed by establishment growth rate? Second, do the high rates of job creation and destruction reported in Tables I and II reflect primarily transitory or persistent establishment-level employment changes? We address these questions in turn.

Gross job creation and destruction are distributed over establishments experiencing the full range of expansion and contraction rates. Figure II displays the distributions of job creation and destruction over this range. The right half of Figure II plots the fraction of job creation accounted for by establishments experienc-



(a) Figure II: Job Creation and Destruction Partitioned by Establishment Growth Rate

TABLE III
PERSISTENCE RATES FOR JOB CREATION AND DESTRUCTION^a

Year ^b (t)	$FPOS_{t1}$	$FPOS_{t2}$	$FNEG_{t1}$	$FNEG_{t2}$
1975	0.73	0.54	0.72	0.62
1976	0.75	0.58	0.79	0.69
1977	0.76	—	0.79	—
1980	0.63	0.43	0.82	0.77
1981	0.60	0.44	0.88	0.82
1982	0.60	—	0.86	—
1985	0.62	—	0.84	—
Simple mean	0.67	0.50	0.81	0.73

a. $FPOS_{it}$ ($FNEG_{it}$) is the fraction of jobs created (destroyed) between March of year $t - 1$ and March of year t that persists through March of year $t + n$.
b. Given the ASM panel structure, the persistence measures can be calculated for all plants only in the indicated years.

(b) Table III: Persistence Rates for Job Creation and Destruction

Read: Figure II partitions job creation and destruction by the magnitude of establishment growth rates. Table III measures how much newly created or destroyed employment persists one and two years later.

Economic message: A large share of job creation and destruction comes from sharp plant-level changes, births, and deaths. The high persistence of job flows implies that these are not mainly temporary layoffs or short-run fluctuations.

10.4 Tables IV and V: Plant Characteristics and Passive Learning

TABLE IV
NET AND GROSS RATES BY TYPE OF PLANT^a

No. of employees	Size ^b					Share ^c
	POS	NEG	NET	SUM	MAX	
1-99	0.140	0.164	-0.023	0.304	0.180	0.246
100-249	0.099	0.120	-0.021	0.219	0.133	0.185
250-499	0.086	0.105	-0.019	0.191	0.120	0.162
500-999	0.070	0.093	-0.023	0.163	0.106	0.134
1000+	0.060	0.078	-0.019	0.138	0.090	0.273
Age in years	Age					Share
	POS	NEG	NET	SUM	MAX	
Births						0.008
1	0.270	0.206	0.064	0.476	0.299	0.018
2	0.169	0.167	0.003	0.336	0.200	0.015
3	0.139	0.117	0.022	0.257	0.148	0.015
4-5	0.133	0.134	-0.001	0.267	0.154	0.045
6-10	0.120	0.121	-0.001	0.240	0.135	0.143
11-14	0.102	0.111	-0.010	0.213	0.123	0.110
15+	0.065	0.097	-0.033	0.162	0.103	0.645
Firm operates:	Ownership type					Share
	POS	NEG	NET	SUM	MAX	
Multiple mfg. plants	0.080	0.103	-0.023	0.184	0.115	0.768
A single mfg. plant	0.131	0.146	-0.016	0.277	0.170	0.232
Census region	Geographic region					Share
	POS	NEG	NET	SUM	MAX	
New England	0.090	0.108	-0.018	0.198	0.122	0.073
Middle Atlantic	0.085	0.121	-0.036	0.205	0.127	0.175
South Atlantic	0.079	0.111	-0.032	0.190	0.126	0.238
E. South Central	0.092	0.107	-0.015	0.198	0.122	0.068
W. South Central	0.092	0.101	-0.009	0.193	0.117	0.154
E. North Central	0.091	0.107	-0.016	0.198	0.124	0.070
W. North Central	0.105	0.115	-0.010	0.220	0.134	0.077
Mountain	0.118	0.114	0.005	0.232	0.138	0.026
Pacific	0.118	0.128	-0.009	0.246	0.149	0.120

a. Figures are size-weighted averages of eleven annual values, except for age group figures. Age group

(a) Table IV: Net and Gross Rates by Type of Plant

Read: Table IV shows that job reallocation is much higher among young, small, and single-unit plants. Table V shows that passive learning explains a substantial share of cross-sectoral variation in reallocation rates, especially across size and ownership groups.

Economic message: Passive learning is useful for explaining why some types of plants have higher reallocation than others, but it is not enough to explain why total job reallocation is so large.

variation in job reallocation rates among two-digit industries and census geographic reasons. Learning explains over half of the variation in job reallocation rates among plants of differing sizes and between single-unit and multi-unit plants.

TABLE V
ESTIMATED FRACTION OF CROSS-SECTORAL VARIATION IN JOB REALLOCATION RATES
DUE TO LEARNING ABOUT INITIAL CONDITIONS

	Sector classification by:			
	Industry	Region	Size	Ownership
Cross-sectoral standard deviation of job reallocation rates	0.042	0.026	0.056	0.050
Fraction of cross-sectoral variance explained by learning about initial conditions, assuming $n = 4$	0.32	0.39	0.51	0.56
Fraction of cross-sectoral variance explained by learning about initial conditions, assuming $n = 6$	0.36	0.48	0.57	0.62

(b) Table V: Estimated Fraction of Cross-Sectoral Variation in Job Reallocation Rates Due to Learning About Initial Conditions

10.5 Tables VI and VII: Between-Group Shifts and Time-Series Variance Decomposition

$$\sum_{s=1} (\text{Job Reallocation in } s - |\text{Net Employment Change in } s|).$$

Table VI reports the results of decomposing excess job reallocation for sectoral classification schemes defined in terms of plant

TABLE VI
FRACTION OF EXCESS JOB REALLOCATION DUE TO BETWEEN-GROUP
EMPLOYMENT SHIFTS

Means of 1978 and 1983 values for total manufacturing								
Group Type	Age	Size	Region	Ownership	2-digit ind.	4-digit ind.	All ex ind. ^b	All ^a
Number	8	5	9	2	20	450	720	14400 ^c
	0.06	0.00	0.00	0.00	0.01	0.12	0.15	0.39
Means of 1978 and 1983 values by two-digit industry								
Group type	Age	Size	Region	Ownership	All ^b			
No. of groups	8	5	9	2	720			
Industry								
Food		0.13	0.01	0.01	0.00		0.36	
Tobacco		0.12	0.03	0.05	0.01		0.62	
Textiles		0.18	0.01	0.01	0.03		0.39	
Apparel		0.20	0.11	0.10	0.02		0.46	
Lumber		0.01	0.00	0.02	0.00		0.36	
Furniture		0.08	0.03	0.04	0.00		0.45	
Paper		0.12	0.08	0.08	0.00		0.43	
Printing		0.08	0.06	0.05	0.00		0.39	
Chemicals		0.06	0.00	0.00	0.00		0.39	
Petroleum		0.26	0.07	0.21	0.00		0.65	
Rubber		0.16	0.12	0.01	0.00		0.48	
Leather		0.12	0.03	0.06	0.01		0.52	
Stone, clay, glass		0.04	0.00	0.02	0.00		0.39	
Primary metals		0.07	0.01	0.05	0.00		0.31	
Fabricated metals		0.05	0.00	0.00	0.00		0.23	
Nonelectric mach.		0.12	0.00	0.00	0.00		0.31	
Electric machinery		0.10	0.00	0.02	0.00		0.33	
Transportation		0.09	0.00	0.01	0.00		0.35	
Instruments		0.11	0.00	0.02	0.00		0.50	
Miscellaneous		0.20	0.05	0.05	0.05		0.55	

(a) Table VI: Fraction of Excess Job Reallocation Due to Between-Group Employment Shifts

state census estimates of construction and impact of aggregate and sectoral effects on job creation and destruction rates.

B. Results

Table VII decomposes the time-series variance of annual job reallocation, creation, and destruction rates using several sectoral classification schemes. According to the first row of the first panel, aggregate and sectoral effects account for 4.2–10.5 percent of the time variation in job reallocation, depending on the classification scheme. Assigning all of the covariance term to the aggregate and sectoral effects, they still account for no more than 20 percent of time variation in annual job reallocation rates. These results show

TABLE VII
DECOMPOSITION OF TIME-SERIES VARIANCE OF JOB REALLOCATION, CREATION, AND
DESTRUCTION

	Sectoral classification scheme						
	Total mfg.	4-digit	2-digit	2-digit, size	2-digit, age	2-digit, owner	2-digit, region
# of sectors	1	450	20	100	40	40	180
Fraction of job reallocation (SUM_r) variance accounted for by							
(a) Sectoral/agg. mean effects	0.03	0.105	0.044	0.044	0.042	0.051	0.053
(b) Idiosyncratic effects	1.026	0.797	0.876	0.816	0.879	0.838	0.917
2cov(a,b)	-0.056	0.098	0.079	0.140	0.078	0.111	0.030
Fraction of job creation (POS_r) variance accounted for by							
(a) Sectoral/agg. mean effects	1.44	1.318	1.395	1.431	1.388	1.459	1.385
(b) Idiosyncratic effects	0.16	0.124	0.136	0.142	0.138	0.149	0.142
2cov(a,b)	-0.60	-0.442	-0.531	-0.573	-0.526	-0.609	-0.526
Fraction of job destruction (NEG_r) variance accounted for by							
(a) Sectoral/agg. mean effects	0.63	0.705	0.658	0.726	0.664	0.680	0.665
(b) Idiosyncratic effects	0.079	0.062	0.068	0.063	0.067	0.066	0.071
2cov(a,b)	-0.025	-0.022	-0.025	-0.022	-0.022	-0.022	-0.022

(b) Table VII: Decomposition of Time-Series Variance of Job Reallocation, Creation, and Destruction

Read: Table VI shows that between-group employment shifts explain little of excess job reallocation. Table VII shows that aggregate and sectoral mean effects explain little of the time-series variation in job reallocation, while idiosyncratic effects explain most of it.

Economic message: Job reallocation is primarily a within-sector and plant-level phenomenon. Cyclical variation in reallocation reflects changes in plant-level dispersion, not merely shifts among industries or regions.

10.6 Tables VIII and IX: Cyclical Behavior and Regression Evidence

TABLE VIII
CYCLICAL BEHAVIOR OF SECTORAL JOB REALLOCATION RATES: TIME-SERIES
CORRELATIONS BETWEEN NET_{st} AND SUM_{st}^{ST}

By alternative sectoral classification schemes							
	Total mfg.	4-digit	2-digit	2-digit, size	2-digit, age	2-digit, owner	2-digit, region
Size-weighted avg. correlation	-0.64	-0.36	-0.55	-0.41	-0.50	-0.45	-0.42
(# < 0)/total	1/1	360/450	20/20	87/100	29/40	31/40	152/177
For particular sector types							
Two-digit industry by							
	Durable	Nondur.	Small	Large	Young	Old	Single Multi
Size-weighted avg. correlation	-0.65	-0.40	-0.20	-0.63	0.06	-0.71	-0.19 -0.53
(# < 0)/total	10/10	10/10	33/40	38/40	9/20	20/20	11/20 20/20

(a) Table VIII: Cyclical Behavior of Sectoral Job Reallocation Rates

REGRESSIONS OF ADJUSTED SECTORAL JOB REALLOCATION RATES ON OWN-SECTOR
AND MANUFACTURING NET GROWTH RATES

Dependent variable in regressions: adjusted sectoral job reallocation rates Summary of regressions and goodness-of-fit measures								
	Sectoral classification scheme							
	2-digit industry	Industry by age	Industry by size	Industry by ownership				
Regression number	(1)	(2)	(3)	(4)				
# of observations	220	440	1100	440				
# of sectoral fixed effects	20	40	100	40				
# of other regressors	2	4	6	4				
Regression R^2	0.78	0.78	0.53	0.75				
Fraction of time-series variation explained by other regressors	0.27	0.08	0.05	0.11				
Estimated responses ($\times 100$) of sectoral job reallocation rates								
Sector type:	Ind.	Young	Old	Small	Med.	Large	Single	Multi
Response to a one st. dev. decline in:								
Mfg. net growth rate	-1.15 (0.12)	-0.13 (0.39)	-1.82 (0.14)	-0.18 (0.60)	-1.51 (0.21)	-1.38 (0.17)	0.20 (0.25)	-1.46 (0.13)
Own net growth rate	-0.24 (0.05)	0.11 (0.19)	-0.26 (0.06)	0.55 (0.79)	-0.13 (0.10)	-0.38 (0.07)	-0.16 (0.10)	-0.23 (0.13)
Based on regression #:	(1)	(2)	(2)	(3)	(3)	(3)	(4)	(4)

a. In regression (1) "other regressors" refers to g_m , the manufacturing net growth rate, and g_o , the own-sector net growth rate deviated about g_m . Relative to regression (1): regression (2) adds interactions of these variables with one age-group dummy; regression (3) adds interactions with two size-class dummies; and regression (4) adds interactions with one ownership-type dummy.
b. The "Fraction of time-series variation explained by other regressors" equals the R^2 from a regression of the deviations about sectoral fixed effects on the time-varying "Other regressors."
c. In computing the estimated responses in the bottom panel, a one standard deviation increase in the own-sector net growth rate is measured as the size-weighted average of the time-series standard deviations of sectoral growth rates. This measure isolates the magnitude of time-series variation in the g_o .
d. Heteroskedasticity-consistent standard errors are in parentheses.

(b) Table IX: Regressions of Adjusted Sectoral Job Reallocation Rates on Own-Sector and Manufacturing Net Growth Rates

Read: Table VIII reports correlations between net job growth and adjusted job reallocation. Table IX estimates how adjusted sectoral reallocation responds to manufacturing and own-sector net growth shocks.

Economic message: Countercyclical reallocation is strongest among old, large, and multi-unit plants. The cycle changes the dispersion of employment changes within sectors, not just the average growth of sectors.

11 Key formulas and intuition

11.1 Why use the symmetric growth rate?

The symmetric growth-rate measure has three advantages. First, it treats births and deaths symmetrically. Second, it is bounded between -2 and 2 , which reduces the influence of extreme conventional growth rates. Third, it maps naturally into job creation and destruction rates when weighted by average establishment size.

11.2 Why does *SUM* matter?

$SUM = POS + NEG$ measures the amount of employment reshuffling. If $POS = 0.10$ and $NEG = 0.10$, net growth is zero but reallocation is 20 percent. A representative-firm model would see no change, but workers and plants experience substantial turnover.

11.3 Why is excess reallocation important?

Excess reallocation is:

$$SUM - |NET|.$$

It captures reallocation beyond the minimum required to achieve the net employment change. High excess reallocation means the economy is not merely expanding or contracting uniformly; it is reshuffling jobs across plants.

12 Main conclusions

12.1 Core conclusion

U.S. manufacturing is characterized by large, persistent, and pervasive gross job flows. Each year, many plants expand while many others contract. This reshuffling is economically large and accounts for a major share of worker mobility.

12.2 Economic intuition

The labor market is not a representative plant. It is a constantly changing distribution of establishments. Aggregate employment growth is the net result of simultaneous job creation and destruction. Recessions are not just downward movements of all plants; they are periods of intensified reallocation and dispersion.

12.3 Incremental contribution revisited

The paper creates the empirical foundation for the modern job-flow literature by defining and measuring gross job creation, gross job destruction, and job reallocation in a large establishment-level data set. It also connects job flows to worker flows and shows that plant-level heterogeneity is essential for macro-labor theory.

12.4 Limitations

The paper focuses on manufacturing and annual March-to-March changes, so it does not capture all sectors or higher-frequency flows. The passive-learning calculation relies on identifying assumptions about when learning is complete and whether non-learning sources of reallocation are age neutral. The paper is primarily descriptive and compositional rather than causal.

12.5 Takeaway

The key takeaway is that gross flows matter. Any serious model of labor-market dynamics, business cycles, or firm evolution must explain not only net employment changes, but also why large amounts of job creation and destruction occur simultaneously within narrowly defined sectors, and why this reallocation intensifies in downturns.

13 One-sentence summary

This note summarizes Davis and Haltiwanger (1992), which uses the Longitudinal Research Datafile on U.S. manufacturing establishments to define and measure gross job creation, gross job destruction, and job reallocation, showing that large and persistent plant-level gross flows underlie net employment changes, account for a substantial share of worker reallocation, and fluctuate counter-cyclically mainly through idiosyncratic changes among older, larger, and multi-unit plants.